

Mott physics, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

- Last appendix
- Figures in introduction

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PHS	Particle-hole symmetry
RMP(s)	Renormalized model parameter(s)
SC	Superconducting
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Appendix A

Superexchange and virtual hopping in Hubbard lattices

Superexchange is a key mechanism in the formation of AF ordering in Hubbard lattices. It is an extension of the direct exchange effect, that we describe in Sec. A.2, that favours antiferromagnetic ordering and takes advantage of the chemical structure of cuprates. The mechanism becomes clear enough if we start with a 2-sites Hubbard toy model.

A.1 Direct exchange and virtual hopping in Hubbard lattices

Consider the 2 sites Hubbard model of Eq. (??),

$$\hat{H} = -t \left(\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\uparrow} + \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\downarrow} + \text{h.c.} \right) + U \left(\hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + \hat{n}_{2\uparrow} \hat{n}_{2\downarrow} \right),$$

with $i = 1, 2$ the site index. For an half-filled system (i.e. with two electrons) the Hilbert space is six-dimensional. We use the notation $|n_{1\uparrow} n_{1\downarrow} n_{2\uparrow} n_{2\downarrow}\rangle$ to indicate the six states of the computational basis:

$$\begin{aligned} |\psi_1\rangle &\equiv |1010\rangle, & |\psi_3\rangle &\equiv |1001\rangle, & |\psi_5\rangle &\equiv |0011\rangle, \\ |\psi_2\rangle &\equiv |1100\rangle, & |\psi_4\rangle &\equiv |0110\rangle, & |\psi_6\rangle &\equiv |0101\rangle. \end{aligned} \quad (\text{A.1})$$

We give a pictorial representation of these states in Fig. A.1. We ordered the states in such a way that the two states with total spin different from zero, $|\psi_1\rangle$ and $|\psi_6\rangle$, are at the extrema of the set.

The matrix elements of the hamiltonian can be computed using this basis,

$$H_{ij} = \langle \psi_i | \hat{H} | \psi_j \rangle \quad \Rightarrow \quad H = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & U & -t & -t & 0 & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & 0 & -t & -t & U & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{A.2})$$

In the above matrix, rows and columns are ordered following the states indices of Eq. (A.1). The states $|\psi_1\rangle$ (both spins up) and $|\psi_6\rangle$ (both spins down) are zero-energy eigenstates. Indeed, they do not interact via Coulomb repulsion (are on different sites) and neither through NNs hopping, due to Pauli exclusion principle.



Figure A.1 | Computational basis of Eq. (A.1) for the 2-sites Hubbard model.

Structure	Eigenstate	Energy
Spin-1/2 singlet	$\frac{ \psi_3\rangle - \psi_4\rangle}{\sqrt{2}}$	$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2}$
Spin-1/2 triplet	$ \psi_1\rangle, \frac{ \psi_3\rangle + \psi_4\rangle}{\sqrt{2}}, \psi_6\rangle$	0
	$\frac{ \psi_2\rangle - \psi_5\rangle}{\sqrt{2}}$	U
	$\frac{ \psi_2\rangle + \psi_5\rangle}{\sqrt{2}}$	$E^+ = \frac{U}{2} + \sqrt{\frac{U^2}{4} + 4t^2}$

Table A.1 | List of exact eigenstates and relative energies for the 2-sites half-filled Hubbard model.

All states with null total spin projection, namely $\{|\psi_i\rangle\}$ for $2 \leq i \leq 5$, give a null matrix elements when computed with the aforementioned states, but non-zero matrix elements among themselves. Note that this set includes both the states with one doubly occupied site and one empty site ($|\psi_2\rangle, |\psi_5\rangle$) and those with two singly occupied sites ($|\psi_3\rangle, |\psi_4\rangle$). These states are connected by the central 4×4 matrix in Eq. (A.2). The latter is diagonalised by the means of a change of basis M ,

$$M \begin{bmatrix} U & -t & -t & \\ -t & & & -t \\ -t & & & -t \\ & -t & -t & U \end{bmatrix} M^\dagger = \begin{bmatrix} E^- & & & \\ & 0 & & \\ & & U & \\ & & & E^+ \end{bmatrix}.$$

where blank slots indicate zeros. In Tab. A.1 we report the eigenvectors and relative eigenvalues obtained from diagonalisation.

The ground-state is the singlet state,

$$\frac{|\psi_3\rangle - |\psi_4\rangle}{\sqrt{2}} = \frac{|1010\rangle - |0101\rangle}{\sqrt{2}} = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}},$$

whose energy is

$$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2} \simeq -\frac{4t^2}{U}.$$

The latter approximation is true if $U \gg t$ (strong coupling limit). The singlet state pairs with a spatially-symmetric (node-less) wavefunction, while triplet states with an antisymmetric wavefunction (which, instead, has a node). The entire triplet (second row of Tab. A.1) has zero energy. Excited states are anti-symmetrised and symmetrised version of the polarised states $|\psi_1\rangle$ and $|\psi_6\rangle$.

A.2 Direct exchange in Hubbard lattices

The fact that we limited to two sites allowed for an exact diagonalisation of the toy hamiltonian of Eq. (??). When moving to lattices, the above procedure cannot be extended easily. By using perturbation theory, instead, we can obtain the same result and easily extend to larger lattices.

Nevertheless, let us obtain the same result of Sec. A.1 by the means of perturbation theory. Consider Eq. (??) separating two parts,

$$\hat{H} = \hat{H}_0 + \hat{H}_1,$$

where

$$\begin{aligned} \hat{H}_0 &\equiv U (\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}), \\ \hat{H}_1 &\equiv -t (\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\uparrow} + \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\downarrow} + \text{h.c.}). \end{aligned}$$



Figure A.2 | Schematisation of a second order virtual hopping. Note that this is a virtual process induced by the kinetic perturbation, not an actual physical process.

At zeroth order in perturbation theory, thus considering just the Coulomb repulsion $\hat{H}_0$, the Hilbert space separates in two manifolds

$$\begin{aligned} \text{Span}(|\psi_1\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_6\rangle) & \quad \text{with} \quad E = 0, \\ \text{Span}(|\psi_2\rangle, |\psi_5\rangle) & \quad \text{with} \quad E = U. \end{aligned}$$

We limit to the ground manifold, thus “excluding from the Hilbert space” the states with double occupation $|\psi_2\rangle, |\psi_5\rangle$.

First order corrections to energy are null:

$$\forall i, j \quad : \quad \langle \psi_i | \hat{H}_1 | \psi_j \rangle = 0,$$

because the hopping operator $\hat{H}_1$ produces a doubly occupied site when applied to $|\psi_3\rangle, |\psi_4\rangle$ and zero when applied to $|\psi_1\rangle, |\psi_6\rangle$ due to Pauli exclusion principle.

We need to move to second order corrections. Let us change basis, and define the singlet state

$$|s\rangle \equiv \frac{|\psi_3\rangle - |\psi_4\rangle}{\sqrt{2}},$$

and the triplet states

$$|t_+\rangle \equiv |\psi_1\rangle, \quad |t_0\rangle \equiv \frac{|\psi_3\rangle + |\psi_4\rangle}{\sqrt{2}}, \quad |t_-\rangle \equiv |\psi_6\rangle.$$

Let also $E_\psi^{(0)}$ be the zeroth-order energy associated to state $|\psi\rangle$. Second order correction to the singlet state energy is given by

$$\begin{aligned} E_s^{(2)} &= \sum_{\psi \neq s} \frac{\langle s | \hat{H}_1 | \psi \rangle \langle \psi | \hat{H}_1 | s \rangle}{E_s^{(0)} - E_\psi^{(0)}} \\ &= \frac{2t^2}{E_s^{(0)} - E_{\psi_2}^{(0)}} + \frac{2t^2}{E_s^{(0)} - E_{\psi_5}^{(0)}} = -\frac{4t^2}{U}. \end{aligned}$$

The above result can be obtained easily by direct computation. Note that the + sign in the middle is obtained as a combination of the internal – sign in the definition of the singlet state, and another – sign arising from Fermi anticommutation rules.

Instead, for the central triplet state

$$\begin{aligned} E_{t_0}^{(2)} &= \sum_{\psi \neq t_0} \frac{\langle t_0 | \hat{H}_1 | \psi \rangle \langle \psi | \hat{H}_1 | t_0 \rangle}{E_{t_0}^{(0)} - E_\psi^{(0)}} \\ &= \frac{2t^2}{E_{t_0}^{(0)} - E_{\psi_2}^{(0)}} - \frac{2t^2}{E_{t_0}^{(0)} - E_{\psi_5}^{(0)}} = 0. \end{aligned}$$

This time, the central – sign is present precisely because of the internal sign of the state $|t_0\rangle$. Finally,

$$E_{t_\pm}^{(2)} = \sum_{\psi \neq t_0} \frac{\langle t_\pm | \hat{H}_1 | \psi \rangle \langle \psi | \hat{H}_1 | t_\pm \rangle}{E_{t_\pm}^{(0)} - E_\psi^{(0)}} = 0,$$

again, because of Pauli exclusion principle.

What makes $|s\rangle$ and $|t_0\rangle$ fundamentally different? The singlet state takes advantage of the second-order virtual process where one spin hops to the neighbouring site and returns back. This

process is schematised in Fig. A.2, and is referred to as “virtual hopping”. But each one of the two electrons can do this. The antisymmetry properties of many-body electronic wavefunctions imply, in the end, that these two virtual processes interfere constructively. For the triplet state $|t_0\rangle$, the two processes are perfectly plausible but interfere destructively. The coupling of electronic spin on neighbouring atoms is known as “direct exchange”.

This perturbative scheme scales up to larger lattices. It can be shown analytically that the low-energy sector of the strong-coupled Hubbard hamiltonian of Eq. (??) is approximated by an effective Heisenberg hamiltonian with antiferromagnetic spin-spin coupling [41].

A.3 Superexchange in cuprates

The scheme of direct exchange explains the insurgence of antiferromagnetism in the Hubbard model at strong coupling, $U \gg t$. The physics of cuprates is slightly more complicated.

Indeed, in the copper oxides active layers, the Cu^{2+} ions are surrounded by oxygens in a closed shell configuration. Hence, copper $d_{x^2-y^2}$ -wave orbitals (where conduction electrons reside) do not overlap. Instead, they overlap with p -wave orbitals of the neighbouring oxygens. This enables a mechanism known as “superexchange”: even if the copper spins cannot interact directly, they use the intermediate p -orbitals as a sort of “bridge” and establish an direct exchange mechanism with the intermediate spins. The resulting interaction, as is “seen” by the copper electrons, is again an effective antiferromagnetic coupling [45].

Appendix B

Gaussian states

In this appendix we discuss the properties of the many-body states we use to approximate the real ground-state of the system within HF method. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

B.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is “to anti-symmetrise” a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow, \downarrow\}} \int d\mathbf{r} \psi_i(\mathbf{r}, \sigma) \psi_j(\mathbf{r}, \sigma) = \delta_{ij} ,$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects and $P \in S_N$ a specific permutation of a given N -uple,

$$P : (x_1, x_2, \dots, x_N) \rightarrow (x_{P(1)}, x_{P(2)}, \dots, x_{P(N)}) .$$

Let F_P be the operator that implements the permutation P on the N variables of a function f ,

$$F_P [f(x_1, x_2, \dots, x_N)] = f(x_{P(1)}, x_{P(2)}, \dots, x_{P(N)}) .$$

We define the anti-symmetrisation operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{P \in S_N} (-1)^{\sigma(P)} F_P[f] ,$$

being $\sigma(P)$ the “parity factor” of the permutation $P \in S_N$. The factor $\sigma(P)$ counts the number of exchanges needed to obtain P : this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1 . By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \dots; \mathbf{r}_N, \sigma_N) = \mathcal{A} \left[\prod_{i=1}^N \psi_i(\mathbf{r}_i, \sigma_i) \right] .$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \dots; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix} .$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \dots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible “Slater-like” many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \rightarrow \hat{c}_i, \hat{c}_i^\dagger.$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^N \hat{c}_i^\dagger |\Omega\rangle, \quad (\text{B.1})$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

B.2 Wick’s theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick’s theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (B.1) for a given set of electronic operators $\hat{c}_i, \hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (B.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \leq i \leq N & \quad \hat{c}_i^\dagger |\Psi\rangle = 0, \\ i > N & \quad \hat{c}_i |\Psi\rangle = 0. \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N, \\ \hat{c}_i & \text{for } i > N. \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the “vacuum” state $|\Psi\rangle$. In other words, any Slater state like Eq. (B.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^D \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \quad \text{where} \quad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C}. \quad (\text{B.2})$$

Consider the product operator:

$$\hat{O} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \dots \hat{O}^{(2k)} \quad \text{where} \quad k \in \mathbb{N}.$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{O} = \dots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \dots \hat{C}_{2k} + \dots,$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j . Now, Wick's theorem states that

$$\begin{aligned}\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k} &= \text{N}\left\{\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\} \\ &+ \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} \text{N}\left\{\prod_{i\neq\alpha,\beta} \hat{C}_i\right\} \\ &+ \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} \text{N}\left\{\prod_{i\neq\alpha,\beta,\gamma,\delta} \hat{C}_i\right\} \\ &+ \cdots,\end{aligned}$$

where $\text{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha\hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \rightarrow \hat{A}_i,$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle\Psi|\text{N}\{\hat{A}_i\hat{A}_j\cdots\}|\Psi\rangle = 0,$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions}),$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. This can be seen as the defining property of gaussian states. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle\hat{c}_\alpha^\dagger\hat{c}_\beta^\dagger\hat{c}_\gamma\hat{c}_\delta\rangle = \langle\hat{c}_\alpha^\dagger\hat{c}_\delta\rangle\langle\hat{c}_\beta^\dagger\hat{c}_\gamma\rangle - \langle\hat{c}_\alpha^\dagger\hat{c}_\beta\rangle\langle\hat{c}_\gamma^\dagger\hat{c}_\delta\rangle,$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

B.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators. In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix},$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M , such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\gamma} \\ \hat{\gamma}^\dagger \end{bmatrix}.$$

The new Nambu vectors $\hat{\gamma}$ and $\hat{\gamma}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}.$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D \times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ (U^\dagger)^2 + (V^\dagger)^2 & U^\dagger U + V^\dagger V \end{bmatrix}, \end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D \times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^D U_{ij} \hat{c}_j + \sum_{j=1}^D V_{ij} \hat{c}_j^\dagger \quad \text{and} \quad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^D U_{ji}^* \hat{c}_j^\dagger + \sum_{j=1}^D V_{ji}^* \hat{c}_j,$$

and substituting in the anti-commutator

$$\begin{aligned} \{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^D U_{ii'} \sum_{j'=1}^D U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^D V_{ii'} \sum_{j'=1}^D V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\ &= \sum_{i'=1}^D U_{ii'} \sum_{j'=1}^D U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^D V_{ii'} \sum_{j'=1}^D V_{j'j}^* \delta_{i'j'} \\ &= [UU^\dagger + VV^\dagger]_{ij} \\ &= \delta_{ij}, \end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0.$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \rangle.$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^D U_{ij} \hat{\gamma}_j + \sum_{j=1}^D U_{ji}^* \hat{\gamma}_j^\dagger \quad \text{and} \quad \hat{c}_i^\dagger \equiv \sum_{j=1}^D V_{ji}^* \hat{\gamma}_j^\dagger + \sum_{j=1}^D V_{ij} \hat{\gamma}_j.$$

In Sec. B.2 we argued that, when averaging a product of operators like the one of Eq. (B.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \rightarrow \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \rightarrow \hat{\gamma}_i, \hat{\gamma}_i^\dagger$.

Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and “see” $|\Psi\rangle$ as a vacuum state.

As in Sec. B.2, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}}. \quad (\text{B.3})$$

The subscripts indicate the name we give to these terms throughout the text. Note that, differently from “pure Slater states” EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov. For simplicity, in this thesis we stick to the name “HF method” referring to this discussion. This argument has been carried out at zero temperature, but can be extended to finite temperature.

Appendix C

Connection with mean-field theories

In this appendix we bridge the gap between HF methods and mean-field theory (MFT). In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations. This is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method we sketched here does not work like that. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [23].

However, there are some points of connection. Let us sketch first the general scheme behind MFT.

C.1 Mean-field theory

Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \quad \text{with} \quad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}.$$

Then we have:

$$\begin{aligned} \hat{A} &= (\langle\hat{B}\rangle + \delta\hat{B}) (\langle\hat{C}\rangle + \delta\hat{C}) \\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle(\hat{C} - \langle\hat{C}\rangle) + (\hat{B} - \langle\hat{B}\rangle)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\ &= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}. \end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are “small”. This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle. \quad (\text{C.1})$$

Within the assumption that independently $\hat{B}$ and $\hat{C}$ fluctuations are “small”, we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are restricting the space of wavefunctions and minimising energy there.

C.2 Hartee-Fock methods

Let us restart from the interacting hamiltonian of Eq. (??). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta} \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\beta} \hat{c}_{\alpha},$$

with $\forall \alpha, \beta: V_{\alpha\beta} = V_{\beta\alpha}^*$. In terms of the general hamiltonian of Eq. (??), we have:

$$B_{\alpha\beta\gamma\delta} = 2V_{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma}. \quad (\text{C.2})$$

We work with HF states. Thus Eq. (??) holds exactly (adapted for two indices instead of four)

$$\alpha \neq \beta \quad : \quad \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle}_{\text{Cooper}}.$$

This gives:

$$\begin{aligned} \langle \hat{V} \rangle &= \sum_{\alpha, \beta} V_{\alpha\beta} \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha \rangle \\ &= \sum_{\alpha, \beta} V_{\alpha\beta} \left(\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle \right), \end{aligned} \quad (\text{C.3})$$

from which we have the equations:

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle} = \delta_{\alpha\beta} \left[\sum_{\gamma} V_{\alpha\gamma} \langle \hat{c}_\gamma^\dagger \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle, \quad (\text{C.4})$$

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle. \quad (\text{C.5})$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + y_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger + y_{\beta\alpha}^* \hat{c}_\alpha \hat{c}_\beta \right] + (\text{some constant}), \quad (\text{C.6})$$

where $x_{\alpha\beta} = x_{\beta\alpha}^*$ to ensure hermiticity. Note that we are actively ignoring terms like $\hat{c}\hat{c}^\dagger$. Anti-commuting these, we get sums of operators like $\hat{c}^\dagger \hat{c}$ and constants. The above expression is the most general. Take its EV:

$$\langle \hat{V} \rangle \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle + y_{\alpha\beta} \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle + y_{\beta\alpha}^* \langle \hat{c}_\alpha \hat{c}_\beta \rangle \right] + (\text{some constant}).$$

Deriving the equation above, we obtain:

$$x_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle} \quad \text{and} \quad y_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle}.$$

Applying Eqs. (C.4) and (C.5), we get the identities:

$$x_{\alpha\beta} = \delta_{\alpha\beta} \left[\sum_{\gamma} V_{\alpha\gamma} \langle \hat{c}_\gamma^\dagger \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle, \quad (\text{C.7})$$

$$y_{\alpha\beta} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle. \quad (\text{C.8})$$

These equations are a rephrasing of Eqs. (??), (??) for the fields $h_{\alpha\beta}^{pq}$, setting $A_{\alpha\beta} = 0$. This can be checked by using Eq. (C.2).

Now: consider the special case where $\forall \alpha: V_{\alpha\alpha} = 0$. This is the case for the EHM. Then the above equations imply:

$$\begin{aligned} \hat{V} &= \sum_{\alpha \neq \beta} V_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha \\ &\simeq \sum_{\alpha \neq \beta} V_{\alpha\beta} \left(\underbrace{\hat{c}_\alpha^\dagger \hat{c}_\alpha \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \hat{c}_\beta^\dagger \hat{c}_\beta}_{\text{Hartree}} - \underbrace{\hat{c}_\alpha^\dagger \hat{c}_\beta \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \hat{c}_\beta^\dagger \hat{c}_\alpha}_{\text{Fock}} \right. \\ &\quad \left. + \underbrace{\hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \hat{c}_\beta \hat{c}_\alpha}_{\text{Cooper}} \right) + (\text{some constant}). \end{aligned}$$

To determine the constant, we take the EV of the above equation. The result should equal Eq. (C.3). Take e.g. the Hartree sector:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle + C = \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle \implies C = -\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle,$$

with C a constant. A similar implication holds for the Fock and the Cooper sectors. Then:

$$\begin{aligned} \hat{V} &= \sum_{\alpha \neq \beta} V_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha & (C.9) \\ &\stackrel{\text{HF}}{\simeq} \sum_{\alpha \neq \beta} V_{\alpha\beta} \left(\hat{c}_\alpha^\dagger \hat{c}_\alpha \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \hat{c}_\beta^\dagger \hat{c}_\beta - \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle \right. & (\text{Hartree}) \\ &\quad - \hat{c}_\alpha^\dagger \hat{c}_\beta \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \hat{c}_\beta^\dagger \hat{c}_\alpha + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle & (\text{Fock}) \\ &\quad \left. + \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle \right). & (\text{Cooper}) \end{aligned}$$

Passing from the first row to the following ones, we decomposed the original quartic operator to a sum of three decomposition much similar to Eq. (C.1). This is the main connection with MFT. In HF theory, we do not write the quartic operator as a product of two quadratic operators and decompose it as we would do in MFT. Instead, a very similar decomposition (summed over the three sectors) is given by the fact that EVs are computed over HF states.

Appendix D

Antiferromagnetism in the HM

In this appendix we give some details about the antiferromagnetic phase in the conventional Hubbard model,

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad \text{where } t, U > 0, \quad (\text{D.1})$$

approximated by HF methods. We use the results discussed in Sec. ??.

D.1 The AF ground-state for the conventional HM

Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix}.$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU.$$

We know from Eq. (??) that the approximated hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H}/U}^{(\text{AF})},$$

where:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix} \quad \hat{s}_{\mathbf{k}\sigma}^{\alpha} \equiv \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix}.$$

Through the unitary matrix $W_{\mathbf{k}\sigma}$ given in Eq. (??), the hamiltonian is mapped to a gas of non-interacting fermions collected in the $\hat{\Gamma}$ spinors of Eq. (??),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix}.$$

The final non-interacting hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + nU\hat{N} - E_{\text{H}/U}^{(\text{AF})},$$

with $E_{\mathbf{k}\sigma}$ the Bogoliubov bands of Eq. (??). In next section we give the explicit form of the many-body ground-state at zero temperature and half-filling. From now on, limitedly to this appendix, we use the simplified notation:

$$\Delta_{\mathbf{k}\uparrow} \rightarrow \Delta,$$

with $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, and $\Delta = mU$.

D.2 Explicit AF ground state at half-filling and $\beta \rightarrow \infty$

Let $|\uparrow_{\mathbf{k}\sigma}\rangle$ and $|\downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the “up” and “down” $\mathbf{k}$ -th pseudo-spin states. Let $|\overline{\downarrow}_{\mathbf{k}\sigma}\rangle$ be the “down” state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle. \quad (\text{D.2})$$

We can link the “down” states via an inverse rotation,

$$\begin{aligned} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y} |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - i \sin \theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} (\hat{s}_{\mathbf{k}\sigma}^+ - \hat{s}_{\mathbf{k}\sigma}^-) |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} |\uparrow_{\mathbf{k}\sigma}\rangle, \end{aligned} \quad (\text{D.3})$$

having we used

$$\hat{s}_{\mathbf{k}\sigma}^{\pm} \equiv \frac{\hat{s}_{\mathbf{k}\sigma}^x \pm i \hat{s}_{\mathbf{k}\sigma}^y}{2}.$$

From Eqs. (??) and (??),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}\sigma}, \quad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}+\pi\sigma}.$$

Evaluating the above operators on the “up” and “down” states, we get

$$\begin{aligned} \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 1, & \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 0, \\ \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 0, & \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 1. \end{aligned}$$

Thus, up to a phase

$$|\uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle, \quad |\downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle,$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k}, \sigma)$ Hilbert subspace vacuum.

It follows that composing Eqs. (D.2) and (D.3) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + e^{2i\zeta_{\mathbf{k}\sigma}} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle, \quad (\text{D.4})$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases, as well as the original sign difference of Eq. (D.3). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= (e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}) \sin \theta_{\mathbf{k}\sigma} \cos \theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma} \sin 2\theta_{\mathbf{k}\sigma}. \end{aligned}$$

From Eq. (??)

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{s}_{\mathbf{k}\sigma}^x | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma}, \end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \implies \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with } k \in \mathbb{Z}.$$

Using Eq. (D.4), we conclude

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle. \quad (\text{D.5})$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (??), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \gamma_{\mathbf{k}\sigma}^{(-)\dagger} |\Omega_{\mathbf{k}\sigma}\rangle. \quad (\text{D.6})$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a π shift in reciprocal space, producing a new free gas where all interactions have been “condensed” in these AF pairs.

D.3 Instability of the commensurate AF phase

The commensurate AF phase is unstable at any finite doping. This result is discussed in [26, 27], and here we recall just the main passages. Consider the local spin operator defined in Eq. (??),

$$\hat{S}_j^\alpha = \frac{1}{2} \hat{\Psi}_j^\dagger \tau^\alpha \hat{\Psi}_j, \quad \hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2}, \quad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}.$$

Let us take a continuum limit, $L_x \rightarrow \infty$ and $L_y \rightarrow \infty$, while the lattice spacing goes to zero.

Continuous SSBs are connected to the presence of a stable gapless Goldstone mode. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the commensurate AF phase, the low-energy collective excitations are transverse spin fluctuations about the antiferromagnetic ground-state, describing SDWs. In order to see if the Goldstone mode is stable for a given doping δ , we need to investigate whether the respective response function diverges at zero frequency. If it does, we have a gapless Goldstone mode and the phase is stable; if not, the phase is unstable. We work at $\omega = 0$ in order to capture gapless modes.

We work in interaction representation, and indicate the spin operators by

$$\hat{\mathbf{S}}(\mathbf{r}, t)$$

The retarded transverse spin-spin response is given by:

$$\chi^{-+}(\mathbf{r}, t; \mathbf{r}', t') \equiv i\text{H}(t - t') \langle \Psi | [\hat{S}^-(\mathbf{r}, t), \hat{S}^+(\mathbf{r}', t')] | \Psi \rangle,$$

being $\text{H}(x)$ the Heaviside function. The argument carried by Singh and Tešanović relies on random phase approximation (RPA). Let χ_0^{-+} be the non-interacting transverse response function. Then

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1 - U\chi_0^{-+}}, \quad (\text{RPA})$$

is the random phase approximation (RPA) proper response function.

We move to momentum-frequency space. Within RPA approximation, low-lying spin waves collective excitations at wavevector $\mathbf{q}$ give a stable gapless Goldstone mode if:

$$1 - U\chi_0^{-+}(\mathbf{q}, 0) = 0. \quad (\text{D.7})$$

Commensurate antiferromagnetism is stable if the above equation is satisfied at $\mathbf{q} \rightarrow \pi$. Indeed, the wavevector π expresses two-sites periodicity in both directions.

Now, as we specified in Sec. ??, commensurate antiferromagnetism is described by the simultaneous SSB:

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1), \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)}. \end{aligned}$$

Thus translational invariance is halved in both directions. The lattice $\mathcal{L}$ is separated in the two sublattices $\mathcal{S}_1, \mathcal{S}_2$ of Fig. ?. As a consequence, the non-interacting response function χ_0^{+-} does not depend strictly on $\mathbf{r}, \mathbf{r}'$ themselves. Instead, it depends only on which sublattice $\mathbf{r}$ and $\mathbf{r}'$ belong to. Because of this χ_0^{+-} is representable as a 2×2 matrix χ_0^{+-} . Then Eq. (D.7) is actually an eigenvalue equation,

$$\mathbb{I} - U\chi_0^{+-}(\mathbf{q}, 0) = \mathbf{0},$$

with $\mathbf{0}$ the 2×2 zero matrix. Diagonalising $\chi_0^{+-}(\mathbf{q}, 0)$ (which is symmetric due to lattice exchange symmetry) we can express the above equation in terms of the diagonal matrix containing its eigenvalues.

Consider the largest eigenvalue $\lambda_\delta(\mathbf{k}, \omega)$ of the matrix $[\chi_0^{+-}]$ at doping δ . We work in momentum-frequency space. The phase is unstable if

$$1 - U\lambda_\delta \neq 0, \quad (\text{D.8})$$

because, as we said, in that case we do not have a pole for the transverse spin-spin response function. Singh and Tešanović [26] found

$$\lambda_0(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) = \frac{1}{U}, \quad (\text{half filling}) \quad (\text{D.9})$$

$$\lambda_\delta(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) = \frac{1}{U} + \frac{2\epsilon_F}{\Delta} N(\epsilon_F). \quad (\text{finite doping}) \quad (\text{D.10})$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. The half-filled AF phase is insulating, because the chemical potential lies within the gap. For an insulator Fermi energy is not defined. Instead, at finite filling the phase is metallic because the chemical potential crosses the upper band.

From Eq. (D.8) and (D.9), we see that the half filled antiferromagnet is stable towards transverse spin fluctuations, because χ^{+-} develops a divergence at zero frequency. At any finite doping, since we are describing a metal, a FS exists and thus $N(\epsilon_F)$ is finite. This makes the phase unstable, and this is the reason we do not discuss the AF phase outside of half-filling.

As we discussed at the beginning of Chap. ??, antiferromagnetism can establish in many ways. For instance, striped antiferromagnets are of particular interest [28]. In order to investigate the most stable physical realisation of an antiferromagnet on the conventional HM, on a pure HF level, we should implement a variational search for the minimum of free energy while varying the AF wavevector $\mathbf{k}$. Notice that at incommensurate wavevectors, $\mathbf{k} \neq \boldsymbol{\pi}$, the unit cell is not as simple as a pair of sites, but can become much larger. Hence, the HF analytical solution would be more complicated than our pseudospin representation of the HM hamiltonian in the commensurate AF channel.

D.4 iHF results

For the conventional HM, antiferromagnetism is described by a single HFP, the magnetisation m . Its self-consistency Eq. (??) reads:

$$m_{i+1} = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]} [f(-E_{\mathbf{k}}[m_i]; \beta, \mu) - f(E_{\mathbf{k}}[m_i]; \beta, \mu)], \quad (\text{D.11})$$

where we indicated as m_i the value of magnetisation at step i . We ran iHF simulations in the ranges:

$$0 \leq U \leq 16, \quad 0 \leq \delta \leq 0.48.$$

We considered also $\delta > 0$, although – as we specified in previous section – a doped commensurate antiferromagnet is actually unstable.

In Fig. D.1 magnetisation m at $\beta = 100$ is plotted. As is visible in Fig. D.1a, ignoring instability issues, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger. Also, for $\delta > 0$, a finite magnetisation m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping).

Fig. D.1b shows the dependence of the magnetisation on doping varying the local repulsion U . Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures.

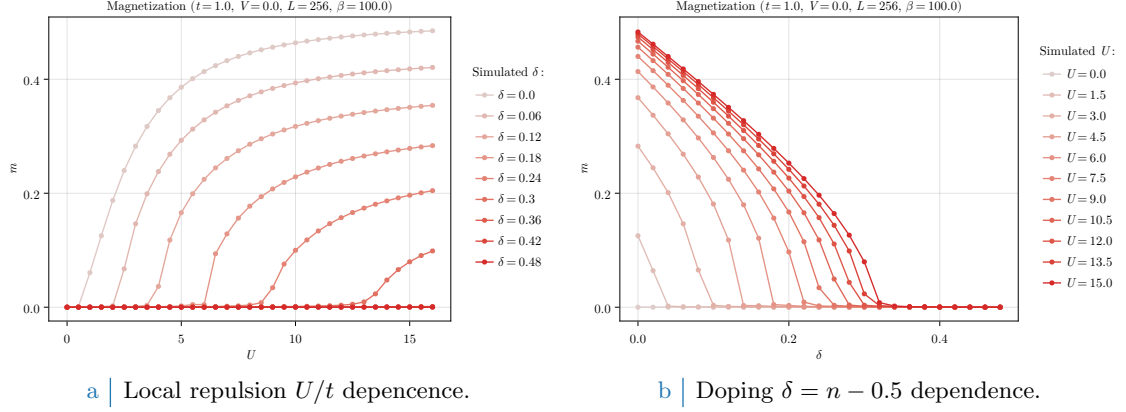


Figure D.1 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. D.1a) and the doping $\delta = n - 1/2$ (Fig. D.1b).

D.5 Analytic expression of self-consistent magnetisation at half-filling

In this subsection we provide an analytic expression to compute m self-consistently at half-filling. Multiply Eq. (??) on both sides by U . Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right),$$

where we used the relation

$$\text{Th}_{\mathbf{k}}^{(-)}[\beta, 0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right).$$

In thermodynamic limit $L \rightarrow \infty$

$$\frac{1}{U} = \frac{1}{2} \int_{\text{BZ}} \frac{d\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right).$$

Substitute the momentum integral with an integral over energies:

$$\frac{2}{U} = \int_{-4t}^{4t} \frac{d\epsilon \rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon^2 + \Delta^2}\right),$$

being $\rho(\epsilon)$ the 2D square lattice DoS. Its explicit form [40] is given by

$$\rho(\epsilon) = \frac{1}{2\pi^2} K\left(\sqrt{1 - \left(\frac{\epsilon}{4t}\right)^2}\right) \quad K(x) \equiv \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - x^2 \sin^2 \theta}},$$

where $K(x)$ is the complete elliptic integral of the first kind, shown in Fig. D.2a.

As is plotted in Fig. D.2b, the resulting DoS is logarithmically peaked at $\epsilon = 0$, on the MBZ. A common approximation is to substitute the actual DoS with a step function,

$$\rho(\epsilon) \simeq \rho_0 H(\epsilon_0 - |\epsilon|) \quad \text{with} \quad 2\rho_0 \epsilon_0 = 1.$$

Here, the peak parameters ϵ_0 and ρ_0 are chosen in order to maintain a correct DoS integral, considering also the 2 pre-factor of spin degeneracy. Within this extremely rough approximation, we have

$$\frac{2}{U} \simeq \rho_0 \int_{-\epsilon_0}^{\epsilon_0} \frac{d\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon^2 + \Delta^2}\right).$$

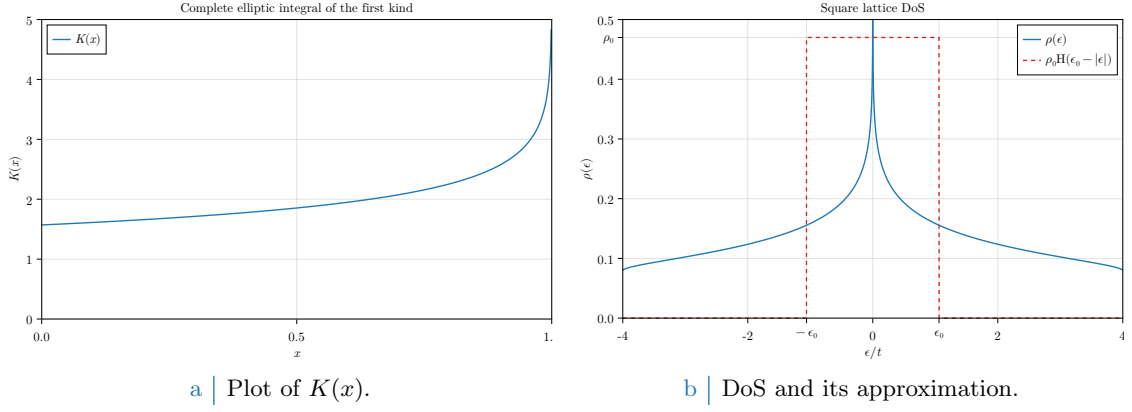


Figure D.2 | Plots of the complete elliptic integral of the first kind $K(x)$ and the density of states for the planar square lattice.

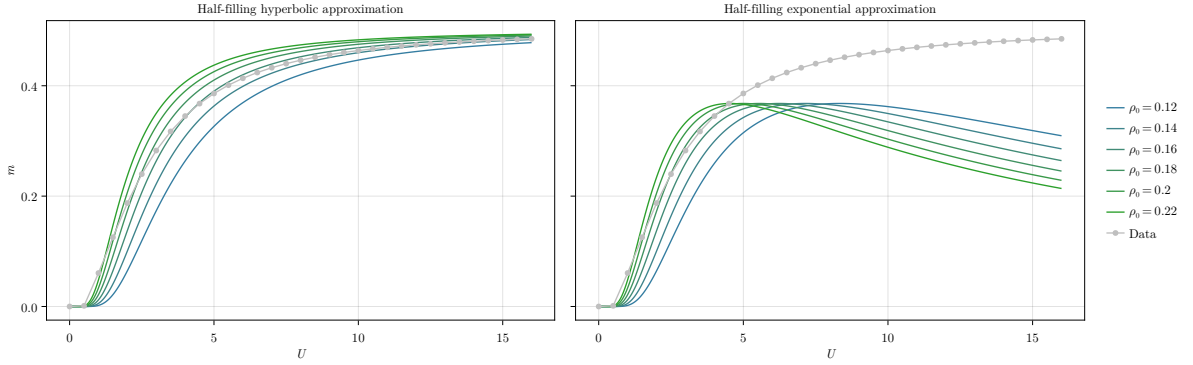


Figure D.3 | Sketch of the analytic expression for m . The left panel shows a family of coloured curves described by Eq. (D.12). The right panel shows a family of coloured curves described by Eq. (D.13), only reasonable at $U \rightarrow 0$.

Suppose the system is magnetised, $\Delta > 0$. At low temperature, $\beta \gg 1$, as a first approximation we can substitute the hyperbolic tangent with unity, obtaining

$$\begin{aligned}
 \frac{2}{\rho_0 U} &\simeq \int_{-\epsilon_0}^{\epsilon_0} \frac{d\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \\
 &= \int_{-\epsilon_0/\Delta}^{\epsilon_0/\Delta} \frac{dx}{\sqrt{1+x^2}} \\
 &= \sinh^{-1} \left(\frac{\epsilon_0}{\Delta} \right) - \sinh^{-1} \left(-\frac{\epsilon_0}{\Delta} \right) \\
 &= 2 \sinh^{-1} \left(\frac{\epsilon_0}{\Delta} \right),
 \end{aligned}$$

thus

$$m \simeq \frac{\epsilon_0}{U \sinh(1/\rho_0 U)}, \quad (\text{D.12})$$

having we used $\Delta = mU$. In the left panel of Fig. D.3 we sketch this function of U varying ρ_0 , ϵ_0 . The same data of Fig. D.1a, $\delta = 0$ are plotted as grey dots. The shown curves are not fits.

Now: at large U ,

$$\lim_{U \rightarrow +\infty} m \simeq \epsilon_0 \rho_0 = \frac{1}{2},$$

which gives the correct saturation limit at high local repulsion. In the opposite limit,

$$\begin{aligned}\lim_{U \rightarrow 0} m &\simeq \lim_{U \rightarrow 0} \frac{2\epsilon_0/U}{\exp(1/\rho_0 U) - \exp(-1/\rho_0 U)} \\ &\simeq \lim_{U \rightarrow 0} \frac{2\epsilon_0}{U} \exp\left(-\frac{1}{\rho_0 U}\right),\end{aligned}\tag{D.13}$$

giving a near-0 exponential suppression of magnetisation, typical of MFT self-consistent solutions. In the right panel of Fig. [D.3](#) we sketch this approximation.

Appendix E

Theoretical constraints on the HFPs

In this appendix we derive some theoretical constraints on the HFPs of the normal and AF phase. The derivations presented here are used in the discussions of Chap. ?? and Chap. ??.

E.1 Normal phase

The normal phase is described by two HFPs:

$$\text{Normal HFPs} \quad : \quad u^{(s^*)}, u^{(d)},$$

which obey the self-consistency equations of Tab. ?. In this section we show that Eq. (??) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0. \quad (\text{E.1})$$

The discussion is divided in three parts: in Sec. E.1.1, we discuss the spatial geometry of bands with a mixed symmetry $s^* \oplus d$. In Sec. E.1.2 we derive Eq. (E.1) by considering the simpler case of unbroken C_4 symmetry (hence setting $u^{(d)} = 0$ “by hand”). In Sec. E.1.3 we break rotational symmetry, $C_4 \xrightarrow{\text{SSB}} C_2$, and terminate the proof.

E.1.1 Mixed $s^* \oplus d$ bands

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$ -wave band. Let $\tilde{\epsilon}_{\mathbf{k}}$ be a generic mixed band,

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y).$$

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^* -wave and d -wave structure factors already presented in Fig. ??, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\}, \quad (\text{E.2})$$

$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\}, \quad (\text{E.3})$$

represented in the top-left panel of Fig. E.1. In the rest of the panels of Fig. E.1 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

$$\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \implies \quad \text{Invert colors in Fig. E.1}$$

$$\epsilon^{(s^*)} > 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \implies \quad \text{Rotate plots by } \pi/2 \text{ in Fig. E.1}$$

$$\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} > 0 \quad \implies \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. E.1}$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\text{MBZ} = \mathcal{A} \cup \mathcal{B} \quad \text{and} \quad \text{NBZ} = \mathcal{A} \cup \mathcal{C},$$

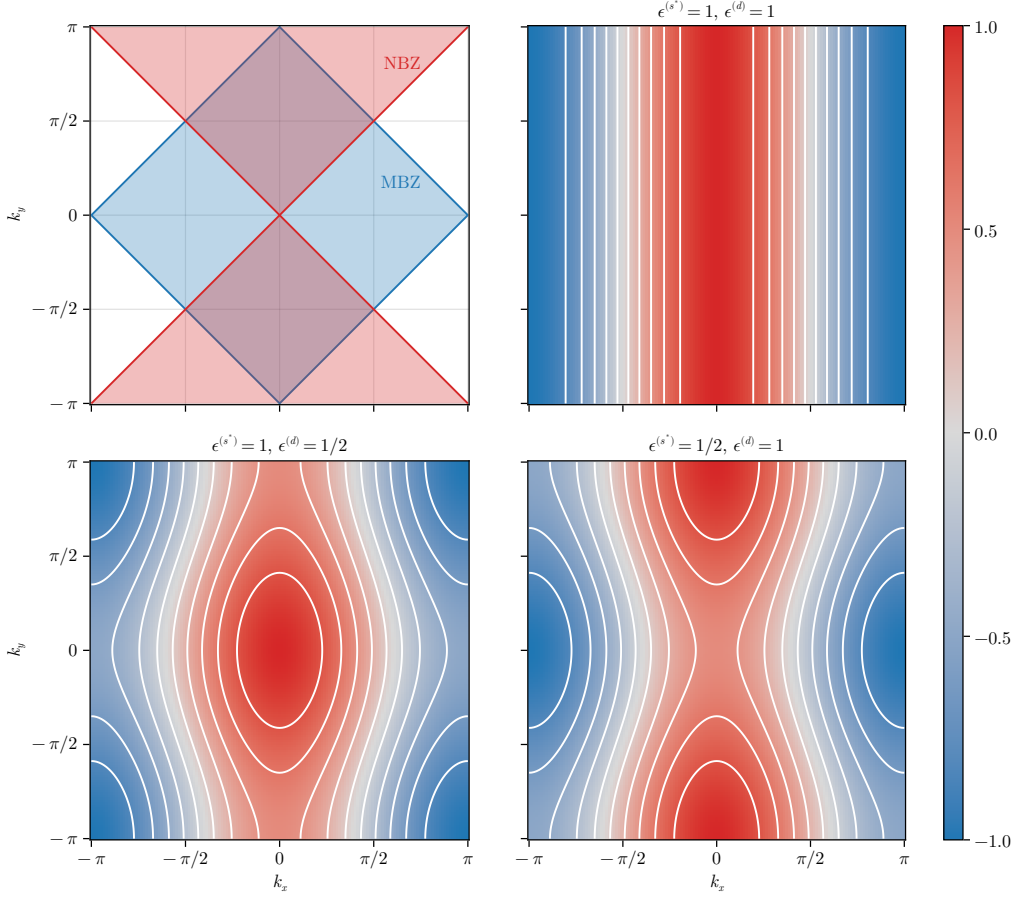


Figure E.1 | Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

being, accordingly to Fig. E.2,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2, \quad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2, \quad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2.$$

These domains satisfy:

$$\mathcal{A} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \quad (\text{E.4})$$

$$= \text{MBZ} \cap \text{NBZ},$$

$$\mathcal{B} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \quad (\text{E.5})$$

$$= \text{MBZ} \setminus \text{NBZ},$$

$$\mathcal{C} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \quad (\text{E.6})$$

$$= \text{NBZ} \setminus \text{MBZ},$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \implies \quad \tilde{\epsilon}_{(k_x, k_y)} = 0,$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. E.2.

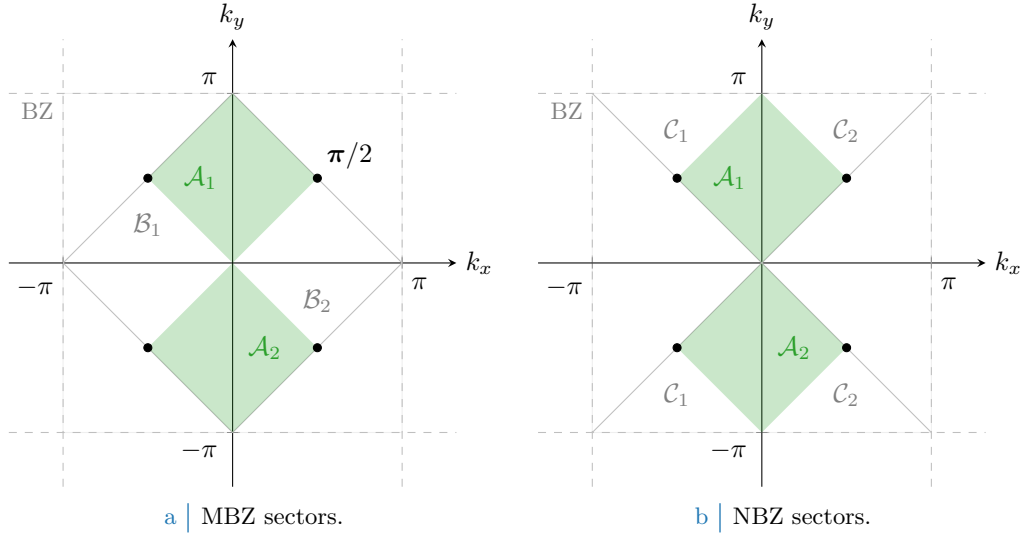


Figure E.2 Depiction of the sectors composing the MBZ (Fig. E.2a) and the NBZ (Fig. E.2b), based on the divisions shown in the top-left panel of Fig. E.1. The full green zone composes $\mathcal{A}$, the region where both the s^* -wave and d -wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^* -wave structure factors is positive-definite while the d -wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

E.1.2 Bands shrinking in C_4

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are given by

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y),$$

with a single self-consistency equation, given by the first row of Tab. ??:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}).$$

We have the following constraint on $u^{(s^*)}$

$$C_4 \implies 0 < u^{(s^*)} < 2t/V. \quad (\text{E.7})$$

The effective hopping is smaller than the original hopping. We can denominate this net effect “bands shrinking”.

Proof E.1 — Proof of Eq. (E.7): bands shrinking effect in C_4 .

The proof follows from this chain of deductions:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$.

Proof. Suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k} \in \text{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k} \in \text{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\pi}],$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu})] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] , \end{aligned} \quad (\text{E.8})$$

having we used the antisymmetry of s^* -wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) ,$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(s^*)} \geq 0 .$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0 .$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$.

Proof. We proceed exactly as did for the previous point, up to Eq. (E.8):

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] .$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) ,$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \implies \quad u^{(s^*)} \leq 0 ,$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V .$$

Collecting the two points above, we obtain the constraint of Eq. (E.7). □

E.1.3 Bands shrinking in $C_2/\mathbb{Z}_2$

Let us extend the arguments above to a weaker symmetry, allowing for

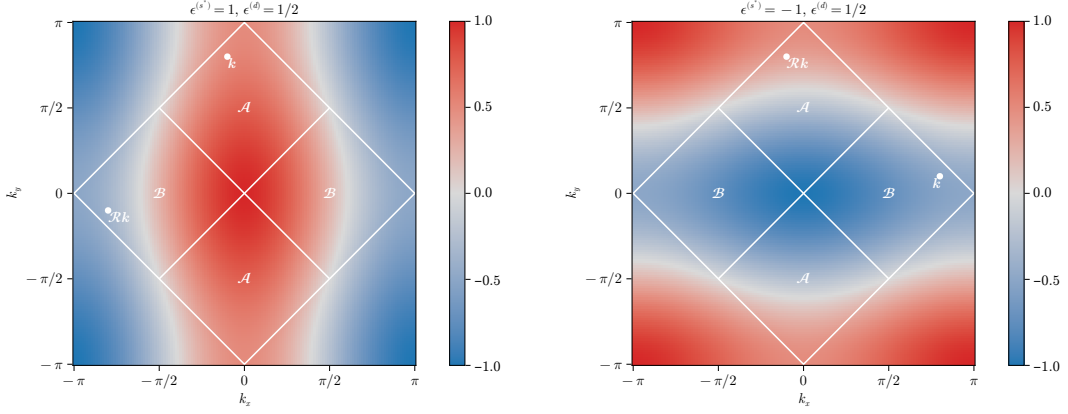
$$C_4 \xrightarrow{\text{SSB}} C_2/\mathbb{Z}_2 .$$

We work with the full, anisotropic band of Eq. (??). In the normal phase, nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 ,$$

which coincides with Eq. (E.1). Therefore, even if we allow for the breaking of planar rotational symmetry, this does not take place.

Proof E.2 — Proof of Eq. (E.1): non-nematic bands shrinking effect in C_2 .



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure E.3 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (E.15), (E.10).

The proof is organised as a series of sub-proofs. We start by proving the left side of Eq. (E.1), namely that

$$0 < u^{(s^*)} < 2t/V.$$

We use everywhere $t > 0$.

1. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V, \quad u^{(d)} > 0 \quad \implies \quad u^{(s^*)} \leq 0. \quad (\text{E.9})$$

Proof. Let us start by the implication

$$u^{(s^*)} \geq 2t/V > 0, \quad u^{(d)} > 0 \quad \implies \quad \epsilon^{(s^*)} \geq 0, \quad \epsilon^{(d)} > 0.$$

In Fig. E.3a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d -wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (E.8), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \\ + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})].$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. E.1, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k}. \quad (\text{E.10})$$

A graphical rendition of this idea is given in Fig. E.3a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \\ \times [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})]. \quad (\text{E.11})$$

From Eq. (E.4), $\forall \mathbf{k} \in \mathcal{A}$ the functions $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0}, \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0}. \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0,$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}), \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}). \end{aligned}$$

This condition, inserted in Eq. (E.11), leads to the absurd of Eq. (E.9).

2. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V > 0, \quad u^{(d)} < 0 \quad \implies \quad u^{(s^*)} \leq 0. \quad (\text{E.12})$$

The proof is identical to the above point, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$.

3. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V > 0 \quad \implies \quad u^{(d)} = 0. \quad (\text{E.13})$$

The proof follows from combining Eqs. (E.9), (E.12).

4. *It is impossible to have $u^{(s^*)} \geq 2t/V$.*

Proof. From Eq. (E.13), we know that if $u^{(s^*)} \geq 2t/V$ then $u^{(d)} = 0$. But then C_4 is unbroken, and we get back to the discussion of Sec. E.1.2. Then we cannot have $u^{(s^*)} \geq 2t/V$.

5. *The following implication holds:*

$$u^{(s^*)} \leq 0, \quad u^{(d)} > 0 \quad \implies \quad u^{(s^*)} > 0. \quad (\text{E.14})$$

Proof. Let for instance

$$u^{(s^*)} \leq 0, \quad u^{(d)} > 0 \quad \implies \quad \epsilon^{(s^*)} < 0, \quad \epsilon^{(d)} > 0.$$

In Fig. E.3b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$. Proceed as for the first point of this chain. Similarly to Eq. (E.10), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k}. \quad (\text{E.15})$$

An equation similar to Eq. (E.11) holds, the sum being performed over $\mathcal{B}$

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \\ &\quad \times [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})]. \end{aligned} \quad (\text{E.16})$$

From Eq. (E.5), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\leq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\leq 0}, \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\leq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\leq 0}. \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0,$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}), \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}). \end{aligned}$$

This condition, inserted in Eq. (E.11), leads to an absurd:

$$u^{(s^*)} \leq 0, \quad u^{(d)} > 0 \quad \implies \quad u^{(s^*)} > 0.$$

which coincides with Eq. (E.14).

6. *The following implication holds:*

$$u^{(s^*)} \leq 0, \quad u^{(d)} < 0 \quad \implies \quad u^{(s^*)} > 0. \quad (\text{E.17})$$

The proof is identical to the above point, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$.

7. *The following implication holds:*

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(d)} = 0. \quad (\text{E.18})$$

The proof follows from combining Eqs. (E.14), (E.17).

8. *It is impossible to have $u^{(s^*)} \leq 0$.*

Proof. From Eq. (E.18), we know that if $u^{(s^*)} \leq 0$ then $u^{(d)} = 0$. But then C_4 is unbroken, and we get back to the discussion of Sec. E.1.2. Then we cannot have $u^{(s^*)} \leq 0$.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (E.7),

$$0 < u^{(s^*)} < 2t/V,$$

holds. In the s^* -wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, “bands shrinking”. What is missing is the behaviour of the d -wave channel when said effect takes place.

We move on and prove the right side of Eq. (E.1), namely that

$$u^{(d)} = 0.$$

Continuing the enumeration from the above points,

9. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \implies \quad u^{(d)} < 0. \quad (\text{E.19})$$

Proof. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. From Tab. ??, the self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{NBZ}} (\cos k_x - \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})], \end{aligned}$$

where we recognised, as is shown in Fig. E.2b, that $\text{BZ} \setminus \text{NBZ} = \mathcal{R}\text{NBZ}$. It holds:

$$\forall \mathbf{k} \in \text{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0}, \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0}, \end{aligned}$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get the absurd of Eq. (E.19).

10. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \implies \quad u^{(d)} > 0. \quad (\text{E.20})$$

The proof is identical to the point above.

11. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \implies \quad u^{(d)} = 0. \quad (\text{E.21})$$

The proof follows from combining Eqs. (E.19), (E.20).

12. *It is impossible to have $u^{(d)} \neq 0$.* The proof follows from combining Eqs. (E.13), (E.18) and (E.21).

Hence we conclude that Eq. (E.1) holds. $\square$

E.2 AF phase

The AF phase divides in two subphases, namely aAF and sAF phases, as reported in Tab. ?. The sAF phase is described by five HFPs,

$$\text{sAF HFPs} \quad : \quad m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)},$$

while for the aAF phase

$$\text{aAF HFPs} \quad : \quad m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}.$$

Self-consistency equations are reported in Eq. (??) and Tabs. ??, ??.

In this section we show that

- Eq. (??) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0. \quad (\text{E.22})$$

Hence also in the AF phase we have a Mott localisation effect;

- Eq. (??) holds,

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0. \quad (\text{E.23})$$

Hence the AF gap is not “complexified”.

The discussion is divided in four parts: in Sec. E.2.1, we derive Eq. (E.22) within C_4 symmetry (as in Sec. E.1.2, setting $u^{(d)} = 0$). In Sec. E.2.2 we prove Eq. (E.23) for the sAF phase within C_2 symmetry. In Sec. E.2.3 we do the same for the aAF phase. We conclude, in Sec. E.2.4, by proving Eq. (E.22) within C_2 symmetry.

E.2.1 Bands shrinking in C_4

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. E.1.2) we can conclude immediately that bands renormalisation is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y).$$

Among the harmonics $u^{(\gamma)}$, just the s^* -wave channel contributes to renormalisation of bands, with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}].$$

We now prove that:

$$0 < u^{(s^*)} < 2t/V. \quad (\text{E.24})$$

This constraint is identical to Eq. (E.7), obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\tilde{H}_V$.

Proof E.3 — Proof of Eq. (E.24).

As we did for the normal phase:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$.

Proof. If $t > 0$ and $u^{(s^*)} \leq 0$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0,$$

the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. This gives

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(s^*)} \geq 0.$$

But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily

$$u^{(s^*)} > 0,$$

which is absurd.

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$.

Proof. If $t > 0$ and $u^{(s^*)} \geq 2t/V$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

the equality only holding on the MBZ contour. Inverting inequalities from point above, we conclude necessarily

$$u^{(s^*)} < 2t/V.$$

which is absurd.

□

E.2.2 Selection rules on the magnetic HFPs in sAF phase

We applied HF methods to the EHM and obtained two flavours of antiferromagnetism, each of which involves five HFPs. The scenario appears to be rich. In this section we present an argument to prove that the actual number of HFPs is much smaller, as many of those are mathematically suppressed.

Let us start from the sAF phase, assuming

$$C_4 \xrightarrow{\text{SSB}} C_2.$$

The self-consistency problem is described by the five HFPs on the first row of Tab. ?? and their self-consistency equations of Tabs. ?? and ??.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}). \quad (\text{E.25})$$

The functional F is linear:

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})], \quad (\text{E.26})$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} : f(\mathbf{k}) \geq 0 \implies F[f(\mathbf{k})] \geq 0. \quad (\text{E.27})$$

The relation above is implied by the fact that both $\tilde{\text{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\text{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}} - \mu)} + 1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}} - \mu)} + 1} \geq 0 \quad \text{and} \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{c}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0,$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (c_x + c_y) \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}}{\tilde{E}_{\mathbf{k}}} [\beta, \tilde{\mu}] \\ &= \frac{[2t - Vu^{(s^*)}]}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2), \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)}V\right] (c_x + c_y) + u_\sigma^{(d)}V(c_x - c_y).$$

In terms of the functional F , we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F[(c_x + c_y)^2] - Vu^{(d)} \times F[c_x^2 - c_y^2] \\ &= \left[2t - Vu^{(s^*)}\right] \times \{F[c_x^2] + 2F[c_x c_y] + F[c_y^2]\} - Vu^{(d)} \times \{F[c_x^2] - F[c_y^2]\}, \end{aligned}$$

having we used the linearity property of Eq. (E.26) in last passage.

Let now:

$$A \equiv F[c_x^2], \quad B \equiv 2F[c_x c_y], \quad C \equiv F[c_y^2], \quad (\text{E.28})$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right] (A + B + C) - Vu^{(d)}(A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)} \right] (A + B + C) - Vu^{(d)}(A - C), \quad (\text{E.29})$$

$$u^{(d)} = \left[2t - Vu^{(s^*)} \right] (A - C) - Vu^{(d)}(A - B + C), \quad (\text{E.30})$$

$$v^{(s^*)} = -Vv^{(s^*)}(A + B + C) - Vv^{(d)}(A - C), \quad (\text{E.31})$$

$$v^{(d)} = -Vv^{(s^*)}(A - C) - Vv^{(d)}(A - B + C). \quad (\text{E.32})$$

These equations define a four-components nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps. Note that the full map would include another equation for the magnetisation. It holds:

$$v^{(s^*)} = 0 \quad \text{and} \quad v^{(d)} = 0. \quad (\text{E.33})$$

Proof E.4 — Proof of Eq. (E.33): all symmetric complexifiers vanish.

Let us start by applying a linear transform on the map: define the new “sum” and “difference” variables

$$S_u \equiv u^{(s^*)} + u^{(d)}, \quad D_u \equiv u^{(s^*)} - u^{(d)}, \quad S_v \equiv v^{(s^*)} + v^{(d)}, \quad D_v \equiv v^{(s^*)} - v^{(d)},$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}.$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (E.29) and (E.30) (the first two) as well as Eq. (E.31) and (E.32) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B), \quad (\text{E.34})$$

$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C), \quad (\text{E.35})$$

$$(1 + 2AV)S_v + BVD_v = 0, \quad (\text{E.36})$$

$$(1 + 2CV)D_v + BVS_v = 0. \quad (\text{E.37})$$

1. *Either $S_v = 0 \wedge D_v = 0$, or $S_v \neq 0 \wedge D_v \neq 0$.*

Proof. Let $S_v = 0$. From Eq. (E.37), using $C = F[c_y^2] \geq 0$ implied by Eq. (E.27):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \implies D_v = 0.$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (E.36) with $C \leftrightarrow A$, we conclude $S_v = 0$. Either both are non-zero or both are zero.

2. *Let $S_v \neq 0 \wedge D_v \neq 0$. Then $B \neq 0$.*

Proof. To prove this, consider for example Eq. (E.36): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V} \frac{S_v}{D_v},$$

which is non-zero because $A = F[c_x^2] \geq 0$, the latter following from Eq. (E.27). We could have used identically Eq. (E.37).

3. Let $S_v \neq 0 \wedge D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV). \quad (\text{E.38})$$

Proof. We multiply Eq. (E.36) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0,$$

and substitute from Eq. (E.37) $BVS_v = -(1 + 2CV)D_v$,

$$[-(1 + 2CV)(1 + 2AV) + (BV)^2] D_v = 0.$$

Since $D_v \neq 0$ by hypothesis, Eq. (E.38) is implied.

4. Let $S_v \neq 0 \wedge D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC). \quad (\text{E.39})$$

Proof. We multiply Eq. (E.34) by BV on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B),$$

and substitute from Eq. (E.35) $BVS_v = -(1 + 2CV)D_v + 2t(B + 2C)$,

$$\begin{aligned} -(1 + 2CV)(1 + 2AV)D_v + 2t(B + 2C)(1 + 2AV) + (BV)^2 D_u &= 2tBV(2A + B) \\ 2t(B + 2C)(1 + 2AV) &= 2tBV(2A + B) \\ B + 2ABV + 2C + 4ACV &= 2ABV + B^2V, \end{aligned}$$

Passing from first to second line we used Eq. (E.38), and in the passage below $t > 0$. Last line implies Eq. (E.39).

5. Let $S_v \neq 0 \wedge D_v \neq 0$. Then $B^2 \neq 4AC$.

Proof. Let $B^2 = 4AC$. Then from Eq. (E.38),

$$(BV)^2 = 1 + 2V(A + C) + 4ACV^2 \implies 1 + 2V(A + C) = 0,$$

which is absurd, being $A \geq 0$, $C \geq 0$ from Eq. (E.27).

6. Using the above point in Eq. (E.39) we get that $V > 0$ satisfies:

$$V = \frac{B + 2C}{B^2 - 4AC}. \quad (\text{E.40})$$

7. Let $S_v \neq 0 \wedge D_v \neq 0$. Then $A = C$.

Proof. We substitute Eq. (E.40) in Eq. (E.38),

$$\begin{aligned} (B^2 - 4AC) \left(\frac{B + 2C}{B^2 - 4AC} \right)^2 - 2 \frac{B + 2C}{B^2 - 4AC} (A + C) - 1 &= 0 \\ (B + 2C)^2 - 2(B + 2C)(A + C) - (B^2 - 4AC) &= 0 \\ B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC &= 0, \end{aligned}$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0.$$

From previous points we know $B \neq 0$. Then $A = C$.

8. *It's impossible to have $S_v \neq 0 \wedge D_v \neq 0$.*

Proof. To prove this, assume the contrary and use the point above to reduce Eq. (E.40) to

$$A = C \implies V = \frac{B + 2C}{B^2 - 4C^2} = \frac{1}{B - 2C}.$$

Inserting the latter in Eq. (E.37),

$$\begin{aligned} \left(1 + \frac{2C}{B - 2C} \right) D_v + \frac{B}{B - 2C} S_v &= 0 \\ \frac{B}{B - 2C} D_v + \frac{B}{B - 2C} S_v &= 0, \end{aligned}$$

which implies

$$D_v + S_v = 0 \implies v^{(s^*)} = 0.$$

Recalling from Tab. ?? the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F[(c_x - c_y)^2],$$

and using the positivity rule of Eq. (E.27) we get

$$V > 0 \quad \text{and} \quad F[(c_x - c_y)^2] \geq 0 \implies v^{(d)} = 0,$$

because no finite positive number can equal any finite negative number, and *viceversa*. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \implies v^{(s^*)} = v^{(d)} = 0 \implies S_v = D_v = 0.$$

Absurd.

9. Combining the above point with our first point, we get that $S_v = 0 \wedge D_v = 0$.

Eq. (E.33) is a consequence of the last point. □

We explored any possibility for S_v, D_v : no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyse similarly the behaviour of the self-consistency equations for $u^{(s^*)}$, $u^{(d)}$. This is done later in Sec. E.2.4. For now, it suffices to note that – limitedly to the sAF phase – gap complexification is not present. This effect is not a consequence of HF derivation, which remains self-coherent and general; it is instead of the peculiar form of the non-linear map defined by the set of self-consistency equations.

E.2.3 Selection rules on the magnetic HFPs in aAF phase

Let us discuss the aAF phase. It is described by the five HFPs on the second row of Tab. ??, as well as the self-consistency equations of Tabs. ?? and ?. We continue using the functional F as defined in Eq. (E.25). We also define A, B, C as in Eq. (E.28). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F[s_x^2], \quad Q \equiv 2F[s_x s_y], \quad R \equiv F[s_y^2]. \quad (\text{E.41})$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = [2t - Vu^{(s^*)}] (A + B + C) - Vu^{(d)}(A - C), \quad (\text{E.42})$$

$$u^{(d)} = [2t - Vu^{(s^*)}] (A - C) - Vu^{(d)}(A - B + C), \quad (\text{E.43})$$

$$v^{(p_x)} = -Vv^{(p_x)}P - Vv^{(p_y)}Q, \quad (\text{E.44})$$

$$v^{(p_y)} = -Vv^{(p_x)}Q - Vv^{(p_y)}R. \quad (\text{E.45})$$

We have:

$$v^{(p_x)} = 0 \quad \text{and} \quad v^{(p_y)} = 0. \quad (\text{E.46})$$

Proof E.5 — Proof of Eq. (E.46): all antisymmetric complexifiers vanish.

Let us write the equation for Q , given in Eq. (E.41), explicitly:

$$\begin{aligned} Q &= F[s_x s_y] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{T}}\text{h}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\text{T}}\text{h}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{k_x = -\pi}^{+\pi} \sin k_x \sum_{k_y = -\pi}^{+\pi} \frac{\tilde{\text{T}}\text{h}_{(k_x, k_y)}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{(k_x, k_y)}} \sin k_y. \end{aligned}$$

Passing from second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function ($\tilde{\text{T}}\text{h}/\tilde{E}$) with $\sin k_y$ integrates to zero. Then

$$Q = 0.$$

Using this result in Eq. (E.44), we get

$$v^{(p_x)} = -Vv^{(p_x)}F[s_x^2].$$

Using the positivity rule of Eq. (E.27) we obtain

$$V > 0 \quad \text{and} \quad F[s_x^2] \geq 0 \quad \implies \quad v^{(p_x)} = 0.$$

Identically, from Eq. (E.45) we get $v^{(p_y)} = 0$. This proves Eq. (E.46). □

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. Eqs. (E.33) and (E.46), combined, give Eq. (E.23). The latter is coherent with the constraint of Eq. (??).

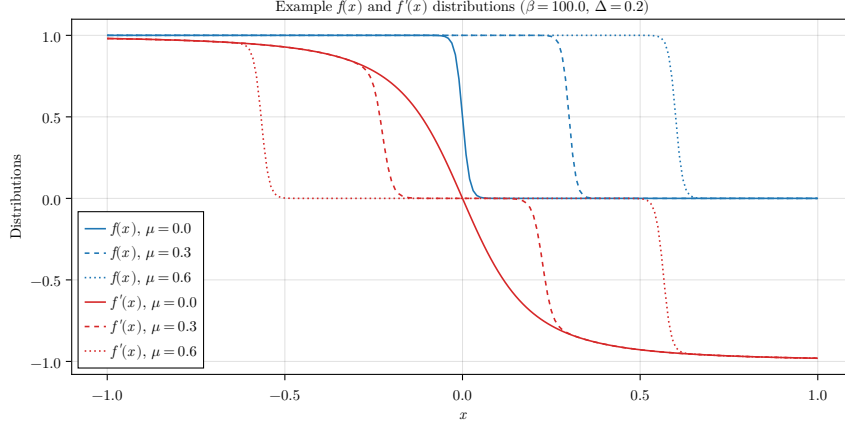


Figure E.4 | Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

E.2.4 Bands shrinking in $C_2/\mathbb{Z}_2$

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \quad \implies \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}.$$

Bogoliubov bands maintain renormalisation through the $u^{(\gamma)}$ HFPs channel. For the AF phase, Eq. (E.22) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0.$$

Starting from a rather complicated scenario, we were able to reduce all the AF HFPs down to the pair

$$m, u^{(s^*)}.$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands.

Proof E.6 — Proof of Eq. (E.22): non-nematic bands shrinking effect in C_2 .

While the proof given in Sec. E.1.3 was long and quite elaborated, this one is shorter. Let us start from recalling the normal phase context:

$$\begin{aligned} \text{Normal phase:} \quad u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}), \\ u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}), \end{aligned}$$

while, using π periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\begin{aligned} \text{AF phase:} \quad u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}], \\ u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]. \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}],$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \rightarrow f'$ exchange. We plot $f(x)$ and $f'(x)$ in Fig. E.4 for various choices of the chemical potential.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. E.1.3 (when showing the same for the normal phase) we never used the specific form of f ; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. E.1.3. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \rightarrow x \quad \text{and} \quad mU \rightarrow \Delta,$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x),$$

with

$$g(x) = -\frac{x}{\sqrt{x^2 + \Delta^2}},$$

$$h(x) = \left(\frac{1}{e^{\beta(\sqrt{x^2 + \Delta^2} - \mu)} + 1} - \frac{1}{e^{\beta(-\sqrt{x^2 + \Delta^2} - \mu)} + 1} \right).$$

The following points are true:

1. The function is antisymmetric about $x = 0$, because $g(x)$ is antisymmetric, while $h(x)$ is symmetric.
2. $g(x)$ is monotonically descendent. Its derivative is given by

$$\begin{aligned} \frac{dg}{dx} &= -\frac{\sqrt{x^2 + \Delta^2} - x^2/\sqrt{x^2 + \Delta^2}}{x^2 + \Delta^2} \\ &= -\frac{\Delta^2}{(x^2 + \Delta^2)^{3/2}} < 0. \end{aligned}$$

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To prove this, we can compute the derivative of f' ,

$$\frac{df'}{dx} = \frac{dg}{dx}h(x) + g(x)\frac{dh}{dx}.$$

We have $h(x) > 0$, and as we said $dg/dx < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2 + \Delta^2} \quad \Rightarrow \quad \frac{dE}{dx} = \frac{x}{E(x)} = -g(x).$$

Then:

$$\begin{aligned} \frac{dh}{dx} &= \frac{d}{dx} \left[\frac{1}{e^{\beta(-E(x) - \mu)} + 1} - \frac{1}{e^{\beta(E(x) - \mu)} + 1} \right] \\ &= \beta \frac{dE}{dx} \underbrace{\left[\frac{e^{\beta(-E(x) - \mu)}}{[e^{\beta(-E(x) - \mu)} + 1]^2} + \frac{e^{\beta(E(x) - \mu)}}{[e^{\beta(E(x) - \mu)} + 1]^2} \right]}_{k(x)}. \end{aligned}$$

The term in square brackets is positive, $k(x) > 0$. Taking all together, we have

$$g(x) \frac{dh}{dx} = -\beta g^2(x) k(x) < 0 \quad \implies \quad \frac{df'}{dx} < 0.$$

Hence f' decreases monotonously.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. [E.1.3](#), and conclude that also for the AF phase

$$u^{(d)} = 0,$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. [\(E.24\)](#).

□

Appendix F

Mixing optimization in HM model

This short appendix is meant to give a practical example on a topic we often referenced to in text: how optimisation of g , the mixing parameter defined in Sec. ?? and given operatively in Eq. (??) through

$$\mathbf{v}_{i+1} \equiv g\mathbf{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i,$$

can be relevant in order to ensure convergence of an iterative HF scheme as presented in Sec. ??.

F.1 Map analysis for s -wave superconductivity in HM

As we discussed in Sec. ??, conventional repulsive HM cannot exhibit BCS s -wave superconductivity within a MFT general scheme. This is proven at the end of Sec. ??, where we end up with the self-consistency equation

$$w^{(s)} = -\frac{U}{U_c[\beta]} w^{(s)}, \quad (\text{F.1})$$

where $U_c[\beta]$ is defined as in Eq. (??),

$$U_c[\beta] \equiv \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\text{Th}_{\mathbf{k}}^{(-)}[\beta]}{E_{\mathbf{k}}} \right]^{-1}. \quad (\text{F.2})$$

Since by construction $U_c[\beta] > 0$ when $U > 0$, Eq. (F.1) can only be solved by $w^{(s)} = 0$. Now, our aim is the following: given any initialiser $w_0^{(s)}$, our iterative HF algorithm should give

$$w_0^{(s)} \xrightarrow{\text{Alg}} \dots \xrightarrow{\text{Alg}} \sim 0.$$

We ask: is this algorithmic evolution possible at any model parametrisation?

F.1.1 Constraints for algorithmic evolution

Consider the nonlinear map of Eq. (F.1) as “seen by the algorithm”: at step i , we compute the left side by using the value of $w^{(s)}$ at step $i-1$ (also for computing $U_c[\beta]$). Let us drop the superscript (s) for the sake of readability. Seen this way, we are effectively moving along the numeric succession

$$w_{i+1} = -g \left(\frac{U}{U_c[\beta]} \right)_i w_i + (1-g)w_i \quad (\text{F.3})$$

the subscript indicating iteration. In previous equation we made use of Eq. (??) to generate the $i+1$ -th step from step i . Let now:

$$U_s \equiv \left(\frac{1}{g} - 1 \right) U_c,$$

$$U_h \equiv \left(\frac{2}{g} - 1 \right) U_c.$$

The subscript indicate a “soft boundary” (first) and an “hard boundary” (second). If $w_i > 0$:

1. It always holds $w_{i+1} \leq w_i$. If instead $w_{i+1} > w_i$, from Eq. (F.3) we have

$$\begin{aligned} -g \left(\frac{U}{U_c[\beta]} \right)_i w_i + (1-g)w_i &> w_i \\ -g \left(\frac{U}{U_c[\beta]} \right)_i w_i - gw_i &> 0, \end{aligned}$$

which is impossible because a sum of negative terms cannot be positive.

2. For $0 < U < U_s$, we have $0 < w_{i+1} < w_i$. Indeed w_{i+1} is greater than zero when:

$$\begin{aligned} -g \left(\frac{U}{U_c} \right)_i w_i + (1-g)w_i &> 0 \\ \frac{1-g}{g} &> \frac{U}{U_c} \\ U_s &> U. \end{aligned}$$

3. For $U_s < U < U_h$, we have $-w_i < w_{i+1} < 0$. From point above, we know $U > U_s$ implies $w_{i+1} < 0$ by simply inverting inequalities. Moreover,

$$\begin{aligned} -g \left(\frac{U}{U_c} \right)_i w_i + (1-g)w_i &> -w_i \\ \frac{2-g}{g} &> \frac{U}{U_c} \\ U_h &> U \end{aligned}$$

4. For $U > U_h$, we have $w_{i+1} < -w_i$. This is proven by inverting inequalities in last point.

If we know U_c , for any given g we can predict the algorithmic image of a given step.

Now, consider U_c as a function of the gap Δ ,

$$U_c(\Delta) = \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{1}{\sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2}} \tanh\left(\frac{\beta}{2} \sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2}\right) \right]^{-1}$$

[To be continued...]

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